

Machine Learning

Session 3

Agenda:

1	Logistic regression
2	Classification metrics

Logistic regression

Logistic regression

Logistic Regression is a supervised learning **classification algorithm** used to predict the **probability of a categorical target variable**. It deals with discrete classes like **Yes or No, 0 or 1, True or False**.

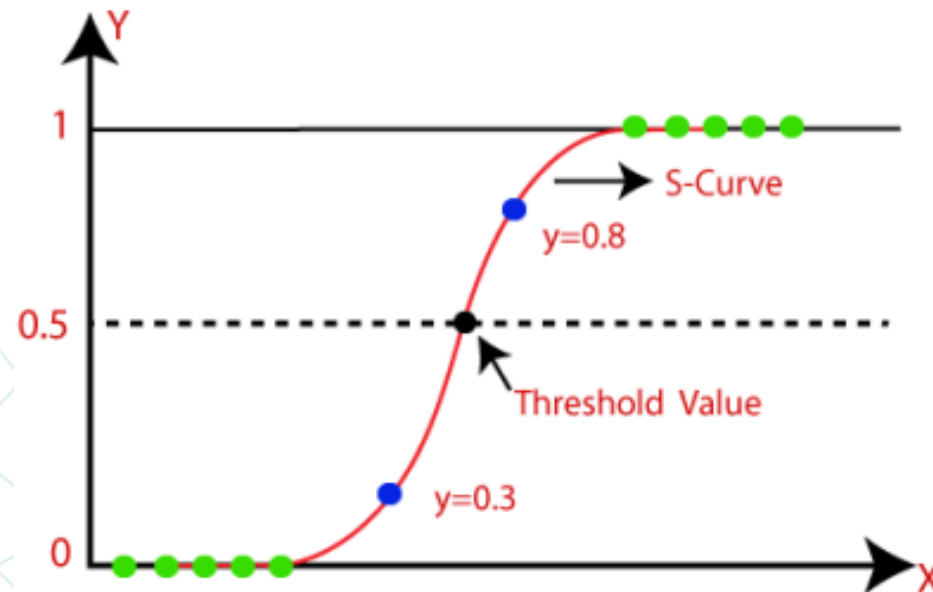
Probabilistic Output

- Unlike linear regression, logistic regression outputs a **probability value between 0 and 1**, indicating the likelihood that a given input belongs to a certain class. Based on a threshold (usually 0.5), logistic regression then classifies the input as either **class 0** or **class 1**.

Logistic regression

S-Curve

- Instead of fitting a straight line (as in linear regression), logistic regression fits an **S-shaped logistic curve**. This sigmoid curve maps the outputs to probabilities between 0 and 1.



Logistic regression

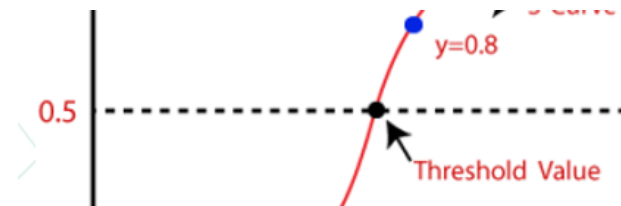
Logistic Function

1. Logistic (Sigmoid) Function

- The logistic function, also known as the **sigmoid function**, is a mathematical function that maps the predicted values to probabilities. The formula is:

$$g(z) = \frac{1}{1 + e^{-z}}$$

- This function produces values between **0 and 1**, making it ideal for classification problems. The curve's shape indicates the decision boundary for classification.



Logistic regression

Logistic Regression Equation

1. Hypothesis for Linear Regression

- The hypothesis of linear regression is given by

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_n x_n = \theta^T x$$

- Here, x represents the input features, and θ represents the coefficients or weights learned by the model.

Logistic regression

Logistic Regression Equation

2. Extension for Logistic Regression

- Logistic regression uses the **same linear hypothesis** but transforms the output using the **sigmoid function** to convert the values into probabilities.

$$h_{\theta}(x) = \theta_0 x_0 + \theta_1 x_1 + \dots + \theta_j x_j = \theta^T x = \begin{bmatrix} \theta_0 & \theta_1 & \dots & \theta_j \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ \dots \\ x_j \end{bmatrix}$$

Where $x_0 = 1$

Logistic regression

Logistic Regression Equation with Sigmoid Function

1. Logistic Hypothesis

- The hypothesis for logistic regression is represented as : $\hat{y} = h_{\theta}(x) = \sigma(\theta^T x)$

Here, σ is the sigmoid function, which converts the linear equation's output into a probability.

Logistic regression

Logistic Regression Equation with Sigmoid Function

2. Probabilistic Interpretation

In this equation, $y^{\wedge}$ represents the predicted probability that the instance x belongs to class 1. Based on this probability, logistic regression makes a decision about the class.

Logistic regression

Sigmoid Function Explanation

1. Application of Sigmoid Function

- The **sigmoid function** is applied to the linear model's raw output to map it to a value between **0 and 1**. This makes it possible to compare the predicted output with a threshold value (e.g., 0.5) and classify the input accordingly.

Logistic regression

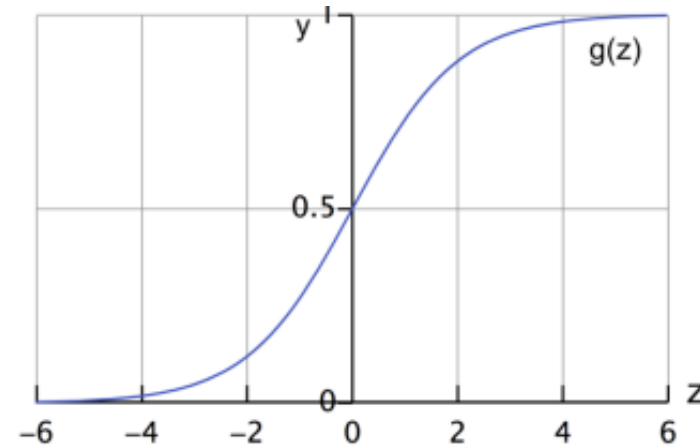
Sigmoid Function Explanation

2. Graph of Sigmoid Function

- The sigmoid function graph shows the curve with values transitioning from **0** to **1**, with the steepest part of the curve near 0.5, indicating the threshold for classification.

Sigmoid Function : $g(z) = \frac{1}{1 + e^{(-z)}}$

Hypothesis : $h_{\theta}(x) = \frac{1}{1 + e^{(-\theta^T x)}}$



Logistic regression

Logistic Regression Decision Boundary

1. Hypothesis for Class 1

- The hypothesis function returns the probability that $y=1$, given x , parameterized by θ . The formula is:

$$h(x) = P(y = 1|x; \theta)$$

2. Decision Boundary

- Logistic regression predicts class 1 if $\theta^T x \geq 0$ or equivalently if $h(x) \geq 0.5$.
- If $\theta^T x < 0$, logistic regression predicts class 0, indicating a probability less than 0.5.

Classification metrics

Classification metrics

When working with classification problems, especially binary or multiclass classification, we need to evaluate how well our model is performing. For this, we use a set of metrics called **classification metrics**.

Classification metrics

1. Confusion Matrix

- The **confusion matrix** is the foundation of most classification metrics. It is a 2x2 table (for binary classification) that shows the **actual vs predicted** values in four categories:
 - **True Positives (TP)**: Correctly predicted positive classes.
 - **True Negatives (TN)**: Correctly predicted negative classes.
 - **False Positives (FP)**: Incorrectly predicted positive classes (also called **Type I Error**).
 - **False Negatives (FN)**: Incorrectly predicted negative classes (also called **Type II Error**).

Classification metrics

1. Confusion Matrix

		Predicted	
		Positive	Negative
Actual	Positive	True positive	False negative
	Negative	False positive	True negative



Classification metrics

2. Accuracy

- Accuracy is the most intuitive and simple classification metric. It measures the percentage of correctly predicted instances out of all instances.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Imagine you have 100 predictions

- 80 are correct (either TP or TN)
- 20 are incorrect (either FP or FN)

Your accuracy would be $\frac{80}{100} = 80\%$

Classification metrics

2. Accuracy

Key Note

- Accuracy is a good metric when the data is **balanced**, meaning there are roughly equal numbers of positive and negative examples.
- However, for **imbalanced datasets**, accuracy can be misleading because the model could simply predict the majority class and achieve high accuracy.

Classification metrics

3. Precision

- **Precision** is the ratio of correctly predicted positive observations to the total predicted positives. It tells us how many of the predicted positives are actually correct.

$$\text{Precision} = \frac{TP}{TP + FP}$$

If your model predicts 30 positives, and 25 of them are actually correct (TP), while 5 are incorrect (FP), the precision would be :

$$\frac{25}{30} = 83.3\%$$

Classification metrics

3. Precision

Key Insight

- Precision is important in situations where **false positives** are costly, such as in email spam classification (false spam predictions are problematic).

Classification metrics

4. Recall (Sensitivity or True Positive Rate)

- **Recall** measures the proportion of actual positives that were correctly identified. It answers the question: **Of all the actual positives, how many did we correctly identify?**

$$\text{Recall} = \frac{TP}{TP + FN}$$

If there are 50 actual positives, and your model correctly predicted 40 of them, but missed 10, your recall would be:

$$\frac{40}{50} = 80\%$$

Classification metrics

4. Recall (Sensitivity or True Positive Rate)

Key Insight

- Recall is crucial in situations where **false negatives** are costly, such as disease detection. Missing a positive case (false negative) is more harmful than incorrectly predicting one.

Classification metrics

5. F1-Score

- The **F1-score** combines precision and recall into one metric by calculating the **harmonic mean** of both. It's particularly useful when you want to balance both precision and recall.

$$F1\text{-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

If your model has a precision of 80% and a recall of 70%, the F1-score would be

$$F1 = 2 \times \frac{0.8 \times 0.7}{0.8 + 0.7} = 74.6\%$$

Classification metrics

5. F1-Score

Key Insight

- The F1-score is a great metric when you need a balance between precision and recall. It's especially useful in imbalanced datasets where accuracy can be misleading.

Classification metrics

6. ROC Curve and AUC (Area Under the Curve)

- The **Receiver Operating Characteristic (ROC) curve** is a graphical representation that shows the performance of a binary classifier as the discrimination threshold is varied.
 - True Positive Rate (TPR) = Recall = $\frac{TP}{TP+FN}$
 - False Positive Rate (FPR) = $\frac{FP}{FP+TN}$

The **ROC curve** plots the TPR against the FPR at different threshold levels. The closer the curve is to the top-left corner, the better the model.

Classification metrics

6. ROC Curve and AUC (Area Under the Curve)

AUC (Area Under the Curve)

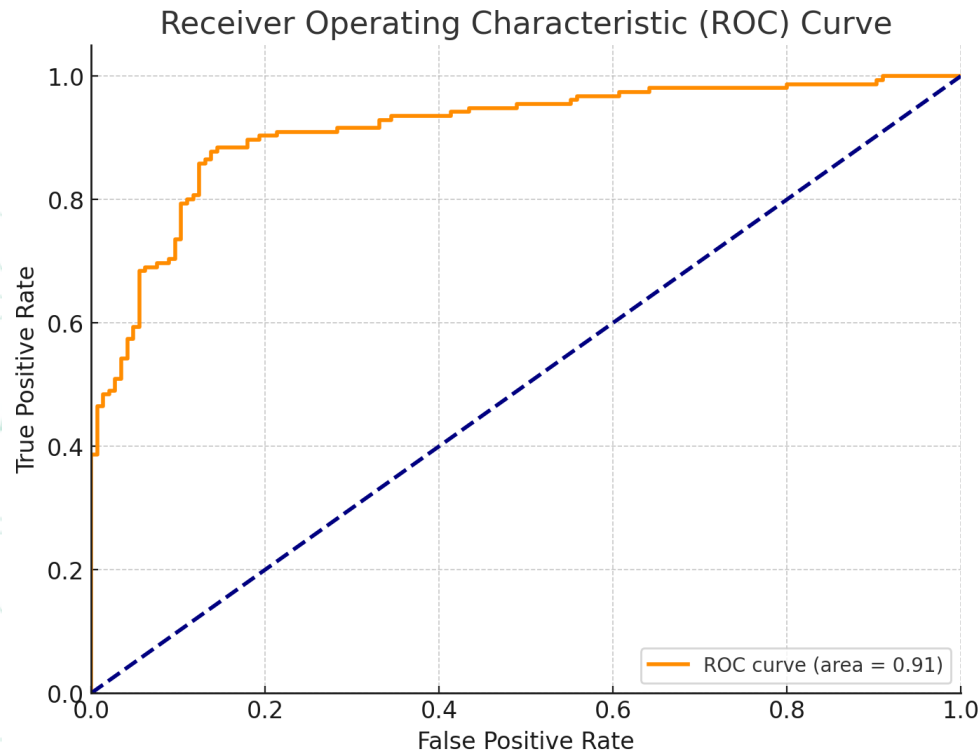
- The **AUC** value represents the degree of separability. Higher AUC means the model is better at distinguishing between positive and negative classes.

AUC ranges from 0 to 1

- 1 = Perfect classifier
- 0.5 = Random guess
- 0 = Completely wrong classification

Classification metrics

AUC (Area Under the Curve)



- ROC Curve is used to evaluate a classification model's performance.
- It plots True Positive Rate (TPR) against False Positive Rate (FPR) at different thresholds.
- The diagonal line represents random guessing.
- The orange line shows the model's performance with an AUC of 0.91, indicating strong performance.
- The closer the curve is to the top left corner, the better the model distinguishes between classes.

Classification metrics

6. ROC Curve and AUC (Area Under the Curve)

Key Insight

- **ROC-AUC** is ideal when you care about the ranking of predictions rather than the exact predicted class.
- It helps to understand how well the model distinguishes between classes across all thresholds.